

Roadmap

Title : AI-Based Data Analytics for Smart Transportation System

Domain : Smart Transportation

Datasets :

I recommend using three benchmark datasets together:

1. Public Transport Dataset (Public transportation analysis and identification of transport-related patterns)

<https://data.opencity.in/dataset/public-transport-data/resource/f482bef5-e59d-4e79-99d9-bf5d643f9b2a>

2. Road Accidents in India Dataset (Road-accident analysis, accident patterns, and safety-related insights)

<https://data.opencity.in/dataset/road-accidents-in-india-2023/resource/3af0e158-4595-4118-b18a-f324662582cf>

3. Climate in India - Daily Weather Dataset (Weather and climate analysis to study environmental factors relevant to transportation)

<https://www.kaggle.com/datasets/developperghost/climate-in-india-daily-weather-data-2000-2024>

All the 3 are famous Valid Datasets. All are to be considered in the Project

Abstract :

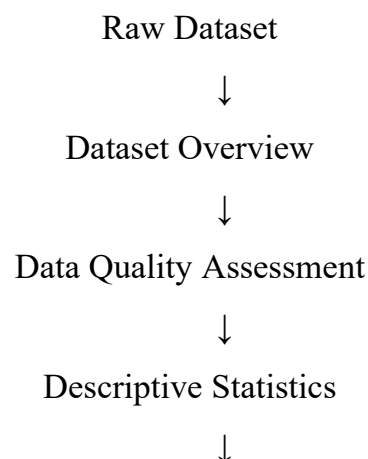
Smart transportation systems generate large volumes of information from public transport services, road-safety records, and environmental conditions. The project AI-Based Data Analytics for Smart Transportation Systems focuses on applying data analytics and artificial intelligence techniques to discover meaningful transportation patterns and support data-driven decision-making. Three datasets are considered in the proposed study: Public Transport Data, Road Accidents in India – 2023, and Climate in India – Daily Weather Data (2000–2024). These datasets provide complementary perspectives on transportation activity, road safety, and weather conditions. The analysis will begin with exploratory data analysis (EDA), including dataset overview, data-quality assessment, descriptive statistics, missing-value analysis, duplicate

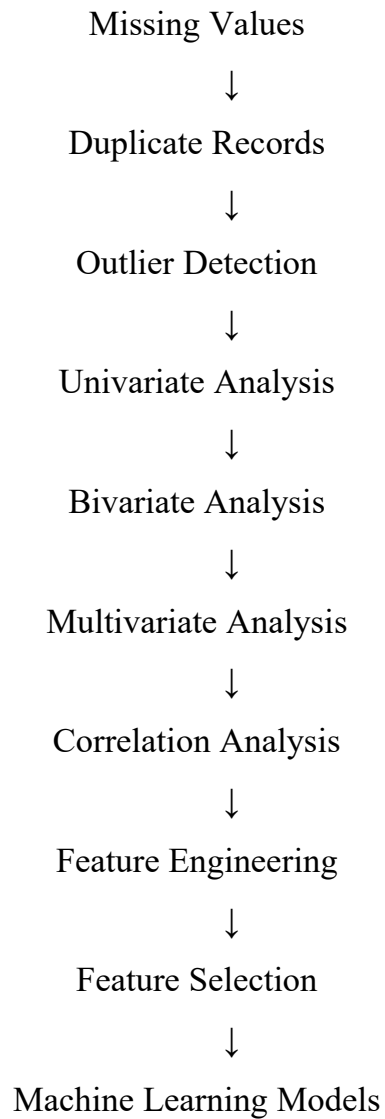
detection, outlier detection, univariate analysis, bivariate analysis, multivariate analysis, correlation analysis, feature engineering, and feature selection. Preprocessing will include missing-value handling, encoding of categorical variables where required, normalization or scaling, and suitable train-test splitting. Machine learning models such as Decision Tree, Random Forest, XGBoost, CatBoost, and Logistic Regression will be considered according to the prediction or classification task formed from the selected datasets. Model performance will be compared using suitable metrics. For regression tasks, MAE, MSE, RMSE, and R^2 can be used; for classification tasks, Accuracy, Precision, Recall, F1-score, and ROC-based analysis can be used. Explainable AI techniques including SHAP and LIME will be incorporated to identify important features and improve the interpretability of model outputs. Visualizations such as heatmaps, boxplots, time-series plots, feature-importance charts, and an interactive dashboard will communicate the results. The project will ultimately provide an integrated analytical workflow for understanding public transport, road accidents, and environmental conditions, with the goal of generating useful insights for smarter and safer transportation planning.

Data Analytics :

- EDA

→ EDA Flow Diagram:





Preprocessing :

- Missing values
- Encoding
- Normalization / Scaling
- Train-Test Split

Machine Learning Models

- Decision Tree
- Random Forest
- XGBoost
- CatBoost

- Logistic Regression (for suitable classification tasks)

Explainability

- SHAP
- LIME

Visualization

- Heatmap
- Boxplot
- Time-Series Analysis
- Feature Importance
- Accident / Transport Pattern Charts
- Interactive Dashboard

Comparative Analysis

Compare Models:

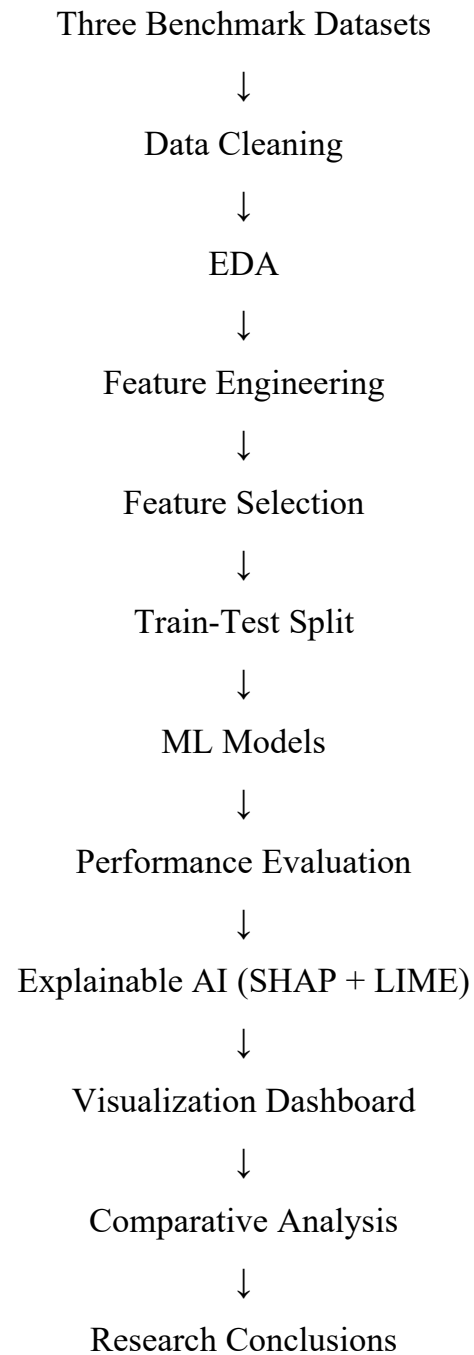
- Decision Tree
- Random Forest
- XGBoost
- CatBoost
- Logistic Regression where applicable

Metrics:

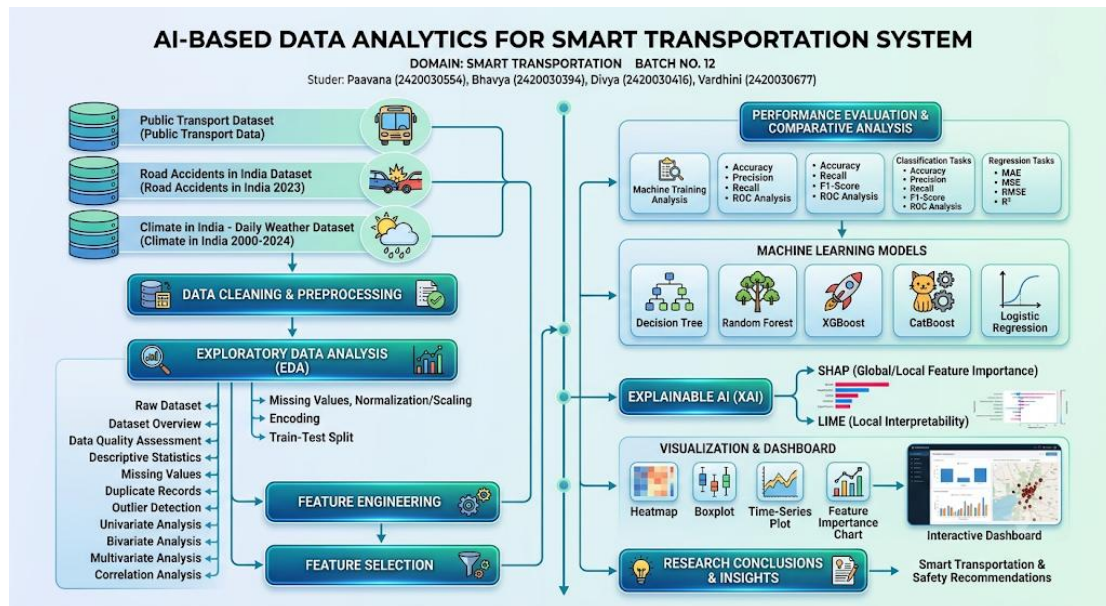
- MAE
- MSE
- RMSE
- R^2
- Accuracy
- Precision
- Recall

- F1-Score
- ROC Analysis

Recommended Research Pipeline :



ARCHITECTURE DIAGRAM :



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